

Lecture 12 – Revision

Question: K–Means Clustering Calculation (15 Marks)

1. Consider the following dataset of 6 points in 2–dimensional space:

Point	x	y
P_1	1	2
P_2	2	2
P_3	1	1
P_4	11	10
P_5	12	11
P_6	10	9

You want to partition these points into **2 clusters** using the k–means algorithm. The initial centroids are:

- Centroid C1: (1, 2)
- Centroid C2: (11, 10)

Perform 2 iterations of calculation and specify whether the algorithm has converged after iteration 2.

Use city–block distance for calculation, e.g., $\mathcal{D}(P_m, P_n) = |P_{m_x} - P_{n_x}| + |P_{m_y} - P_{n_y}|$.

 **Solution**

Question: Naive Bayes Classification (15 Marks)

2. A bank is building a simple fraud–detection system to classify whether a credit–card transaction is **Fraud = Yes/No** using two binary features. The training dataset is:

Transaction	Feature 1 (International?)	Feature 2 (High Amount?)	Fraud
1	Yes	Yes	Yes
2	Yes	No	Yes
3	No	Yes	Yes
4	No	No	No
5	No	No	No
6	Yes	Yes	Yes
7	No	Yes	No
8	Yes	No	No

A new transaction has **International = Yes** and **High Amount = Yes**. Using the **Naive Bayes classifier** (no smoothing unless stated), predict whether the transaction is **Fraud**. Show your calculation of the prior and likelihood terms.

 **Solution**

Question: Confusion Matrix & Classification Metrics (15 Marks)

3. A binary classifier is used to detect whether a chest X-ray indicates **Pneumonia (Positive)** or **No Pneumonia (Negative)**. On a test set of **250** X-rays, the model produced the following outcomes:

- True Positives (TP) = 60
- False Positives (FP) = 15
- False Negatives (FN) = 25
- True Negatives (TN) = 150

Tasks:

1. Construct the **confusion matrix** (Actual vs Predicted) and clearly label TP, FP, FN, TN. (4 Marks)
2. Compute the following metrics and show your working: (4 Marks)
 - Accuracy: Overall proportion correct: $(TP + TN) / (TP + FP + FN + TN)$.
 - Precision: Of predicted positives, the fraction that are truly positive: $TP / (TP + FP)$.
 - Recall (Sensitivity): Of actual positives, the fraction correctly found: $TP / (TP + FN)$.
 - Specificity: Of actual negatives, the fraction correctly rejected: $TN / (TN + FP)$.
3. Compute the **F1-score**: Harmonic mean of precision and recall: $2PR / (P + R)$. (2 Marks)

4. If the decision threshold is increased and **FP decreases by 8** while **FN increases by 8** (assume total stays 250), recompute **precision** and **recall**, and state which one improved and which one worsened. (5 Marks)

☰ Solution

Question: Hierarchical Clustering Calculation (15 Marks)

4. Consider the following dataset of 5 points in 2-dimensional space:

Point	x	y
A	1	1
B	2	3
C	4	2
D	7	6
E	9	7

Perform **agglomerative hierarchical clustering** using **single linkage** (minimum distance) until only 2 clusters remain.

Use city-block distance for calculation, e.g., $\mathcal{D}(P_i, P_j) = |P_{i_x} - P_{j_x}| + |P_{i_y} - P_{j_y}|$.

Draw the dendrogram showing the hierarchical structure and determine 2 clusters according to similarity between samples.

☰ Solution

Question: Likelihood Function

5. You toss a coin **8 times** and observe **3 Heads** and **5 Tails**. Assume each toss is independent and the probability of Heads is p .

Write the **likelihood function** $L(p)$ for this dataset.

Compute the likelihood values $L(0.4)$ and $L(0.6)$, and state which value of p better explains the observed data.

☰ Solution

Question: Correlation Analysis

A researcher wants to study the relationship between weekly study time and exam score for 8 students. The data are:

- Study time x (hours): 1, 2, 3, 5, 5, 6, 7, 9

- Exam score y (out of 100): 50, 55, 58, 66, 69, 73, 76, 85
1. Calculate the correlation between x and y with Pearson's correlation coefficient. The formula for Pearson's correlation coefficient is:
$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$
 2. Based on your result, do you agree the following statement? "Increasing study time causes higher exam scores."

Solution

Short Answer Questions:

- Review GMM and SVM.